

# Dongfang Zhao, Ph.D.

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🌐 <https://hpdic.github.io/>



## Employment

- 2017 – now     📌 **Assistant Professor**, University of Nevada, Reno.
- 2018 – now     📌 **Lead Developer**, Moka Blox LLC, Sacramento, CA.
- 2018 – 2020     📌 **Visiting Professor**, University of California, Davis.
- 2016 – 2017     📌 **Research Associate**, University of Washington, Seattle.
- 2015     📌 **Research Intern**, IBM Almaden Research Center, San Jose, CA.
- 2014     📌 **Research Intern**, Argonne National Laboratory, Chicago, IL.
- 2013     📌 **Research Intern**, Pacific Northwest National Laboratory, Seattle, WA.
- 2009 – 2011     📌 **Software Developer**, Epic Systems Corporation, Madison, WI.
- 2008 – 2009     📌 **Software Developer**, IDMI Systems, Atlanta, GA.

## Education

- 2005     📌 **Diploma (equivalent to U.S. Master), Katholieke Universiteit Leuven (Belgium)** in Statistics  
Thesis Advisor: *Irène Gijbels*
- 2008     📌 **M.Sc., Emory University** in Computer Science
- 2015     📌 **Ph.D., Illinois Institute of Technology** in Computer Science  
Dissertation Committee: *Ioan Raicu (advisor), Xian-He Sun, Zhiling Lan*
- 2016     📌 **Sloan Postdoctoral Fellow, University of Washington** in Computer Science and Engineering  
Mentor: *Magdalena Balazińska*

## Selected Publications

### Journal Articles

- 1 Wang, X., Xu, C., Wang, K., Yan, F., & Zhao, D. (2020). Memory scaling of cloud-based big data systems: A hybrid approach. *IEEE Transactions on Big Data (TBDATA)*. doi:10.1109/TBDATA.2020.3035522
- 2 Zhao, D., Mohamed, M., & Ludwig, H. (2020). Locality-aware scheduling for containers in cloud computing. *IEEE Transactions on Cloud Computing (TCC)*, 8(2), 635–646. doi:10.1109/TCC.2018.2794344
- 3 Zhao, D., Qiao, K., Zhou, Z., Li, T., Lu, Z., & Xu, X. (2017). Toward efficient and flexible metadata indexing of big data systems. *IEEE Transactions on Big Data (TBDATA)*, 3(1), 107–117. doi:10.1109/TBDATA.2016.2640301
- 4 Zhao, D., Liu, N., Kimpe, D., Ross, R. B., Sun, X., & Raicu, I. (2016). Towards exploring data-intensive scientific applications at extreme scales through systems and simulations. *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 27(6), 1824–1837. doi:10.1109/TPDS.2015.2456896
- 5 Zhao, D., Qiao, K., Yin, J., & Raicu, I. (2016). Dynamic virtual chunks: On supporting efficient accesses to compressed scientific data. *IEEE Transactions on Service Computing (TSC)*, 9(1), 96–109. doi:10.1109/TSC.2015.2456889
- 6 Zhao, D., Wang, K., Qiao, K., Li, T., Sadooghi, I., & Raicu, I. (2016). Toward high-performance key-value stores through GPU encoding and locality-aware encoding. *Journal of Parallel and Distributed Computing (JPDC)*, 96, 27–37. doi:10.1016/j.jpdc.2016.04.015
- 7 Zhou, Z., Yang, X., Zhao, D., Rich, P., Tang, W., Wang, J., & Lan, Z. (2016). I/o-aware bandwidth allocation for petascale computing systems. *Parallel Computing (PARCO)*, 58, 107–116. doi:10.1016/j.parco.2016.05.005
- 8 Zhao, D., & Yang, L. (2009). Incremental isometric embedding of high-dimensional data using connected neighborhood graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 31(1), 86–98. doi:10.1109/TPAMI.2008.34

## Conference Proceedings

- 1 Al-Mamun, A., Shen, H., & Zhao, D. (2022). Dean: A lightweight and resource-efficient blockchain protocol for reliable edge computing. In *36th IEEE International Parallel and Distributed Processing Symposium (IPDPS)*.
- 2 Tawose, O. T., Li, B., Yang, L., Yan, F., & Zhao, D. (2022). Topological modeling and parallelization of multidimensional data on microelectrode arrays. In *36th IEEE International Parallel and Distributed Processing Symposium (IPDPS)*.
- 3 Al-Mamun, A., Yan, F., & Zhao, D. (2021a). BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*.
- 4 Al-Mamun, A., Yan, F., & Zhao, D. (2021b). Scichain: Blockchain-enabled lightweight and efficient data provenance for reproducible scientific computing. In *37th IEEE International Conference on Data Engineering (ICDE)*.
- 5 Wang, X., Tawose, O. T., Yan, F., & Zhao, D. (2020). HDK: toward high-performance deep-learning-based kirchhoff analysis. In *The 34th AAAI Conference on Artificial Intelligence (AAAI)*.
- 6 Qin, H., Zawad, S., Zhou, Y., Yang, L., Zhao, D., & Yan, F. (2019). Swift machine learning model serving scheduling: A region based reinforcement learning approach. In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*.
- 7 Wang, J., Al-Mamun, A., Li, T., Jiang, L., & Zhao, D. (2018). Toward performant and energy-efficient queries in three-tier wireless sensor networks. In *The 47th International Conference on Parallel Processing (ICPP)*.
- 8 Mehta, P., Dorkenwald, S., Zhao, D., Kaftan, T., Cheung, A., Balazinska, M., ... AlSayyad, Y. (2017). Comparative evaluation of big-data systems on scientific image analytics workloads. In *Proceedings of the VLDB Endowment*.
- 9 Zhou, Z., Yang, X., Zhao, D., Rich, P., Tang, W., Wang, J., & Lan, Z. (2015). I/o-aware batch scheduling for petascale computing systems. In *2015 IEEE International Conference on Cluster Computing (CLUSTER)*.
- 10 Li, T., Zhou, X., Brandstatter, K., Zhao, D., Wang, K., Rajendran, A., ... Raicu, I. (2013). ZHT: A light-weight reliable persistent dynamic scalable zero-hop distributed hash table. In *27th IEEE International Symposium on Parallel and Distributed Processing (IPDPS)*.
- 11 Zhao, D., Shou, C., Malik, T., & Raicu, I. (2013). Distributed data provenance for large-scale data-intensive computing. In *2013 IEEE International Conference on Cluster Computing (CLUSTER)*.

## Selected Awards

|      |                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | ■ <b>Research Grant</b> , \$250K, National Science Foundation (STTR).                                                                                                                        |
| 2020 | ■ <b>Research Grant</b> , \$240K, U.S. Department of Energy (SBIR).                                                                                                                          |
| 2019 | ■ <b>Best Paper Award</b> , International Conference on Cloud Computing.<br>■ <b>Research Grant</b> , \$30K, Amazon Web Services.<br>■ <b>Research Grant</b> , \$15K, Google Cloud Platform. |
| 2018 | ■ <b>Best Student Paper Award</b> , IEEE International Conference on Cloud Computing.                                                                                                        |
| 2017 | ■ <b>Research Grant</b> , \$20K, Microsoft Azure.<br>■ <b>Research Grant</b> , \$10K, Amazon Web Services.                                                                                   |
| 2016 | ■ <b>Fellowship</b> , \$160K, Sloan Foundation and Washington Research Foundation (WRF).                                                                                                     |
| 2007 | ■ <b>Fellowship</b> , \$85K, Oak Ridge Institute for Science and Education (ORISE).                                                                                                          |

## Selected Services

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|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Journal Editorial Board | ■ IEEE Transactions on Parallel and Distributed Systems (2021–now).<br>■ International Journal of Cooperative Information Systems (2017–now). |
| Conference Committee    | ■ IEEE International Conference on Big Data (2020, 2021).<br>■ ACM/IFIP International Middleware Conference (2019).                           |
| Proposal Review Panel   | ■ National Science Foundation: CISE Division (2018).<br>■ U.S. Department of Energy: Office of Science (2018).                                |